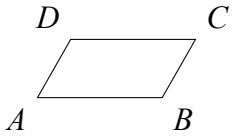
# 1 Vectors I

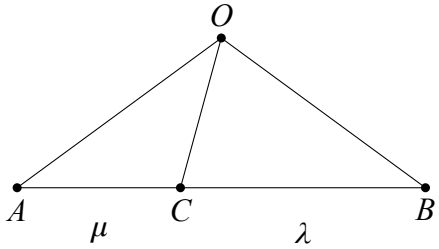
**DRAFT:** Working draft. All examples are taken from your Vectors Summary Notes; the workings are mine, checked with sympy.

## 1.1 Basic Formulas

| | Formula | Note |
|---|---|---|
| Magnitude | $\lvert\mathbf{a}\rvert = \sqrt{x^2+y^2+z^2}$ | |
| Unit vector | $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$ | $\lvert\hat{\mathbf{a}}\rvert = 1$ |
| Parallel vectors | $\mathbf{a} = k\mathbf{b}$ $\quad$ $\mathbf{a}\times\mathbf{b} = \mathbf{0}$ | $k$ scalar constant |
| Collinear $A, B, C$ | $\overrightarrow{AB} \parallel \overrightarrow{BC}$ (or any two of $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{AC}$) | Try drawing the diagram and take any two vectors, to show that they are parallel. |
| Parallelogram $ABCD$ | $\overrightarrow{AB} = \overrightarrow{DC}$, $\quad$ $\overrightarrow{AD} = \overrightarrow{BC}$ |   |

### The Ratio Theorem

Let the points $A$, $B$ and $C$ have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, and suppose that $\lambda AC = \mu CB$. Then the position vector of $C$ is given by the formula

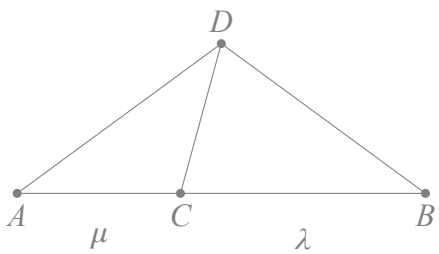



**Ratio Theorem**

$$\mathbf{c} = \frac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$$

$AC : CB = \mu : \lambda$, so $C$ divides $AB$ with $\mu$ beside $A$ and $\lambda$ beside $B$.

The same theorem, starting from a point $D$ instead of $O$:

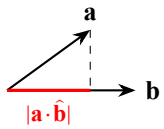
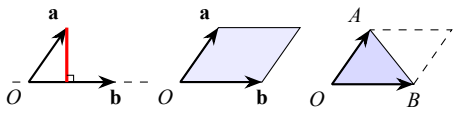



**Ratio Theorem**

$$\overrightarrow{DC} = \frac{\lambda\overrightarrow{DA} + \mu\overrightarrow{DB}}{\lambda + \mu}$$

# 1.2 Scalar Product and Vector Product

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}, \qquad \mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}, \qquad \theta=\text{non-reflex } \angle AOB$$

| | Scalar Product (Dot) | Vector Product (Cross) |
|---|---|---|
| Definition | $\mathbf{a}\cdot\mathbf{b}=a_1b_1+a_2b_2+a_3b_3$ | $\mathbf{a}\times\mathbf{b}=\begin{pmatrix} a_2b_3-a_3b_2 \\ -(a_1b_3-a_3b_1) \\ a_1b_2-a_2b_1 \end{pmatrix}$ |
| Result | scalar | vector, $\perp$ to both $\mathbf{a}$ and $\mathbf{b}$ |
| Formula | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ | $\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$ |
| Angle | $\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ | $\sin\theta=\dfrac{\lvert\mathbf{a}\times\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$<br>this method only finds the basic angle possible, but usually undesirable |
| Properties | $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$<br>$\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$<br>$\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$<br>$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$ | $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$<br>$\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$<br>$\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$<br>$\mathbf{a}\times\mathbf{a}=\mathbf{0}$ |
| Test | $\mathbf{a}\cdot\mathbf{b}=0 \iff \mathbf{a}\perp\mathbf{b}$<br>$>0$ acute, $<0$ obtuse, $=0$ right angle | $\mathbf{a}\times\mathbf{b}=\mathbf{0} \iff \mathbf{a}\parallel\mathbf{b}$<br>(nonzero $\mathbf{a}$, $\mathbf{b}$) |
| Applications | Length of projection $=\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$<br>Projection vector $=(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$ | Perpendicular distance $=\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$<br>Area of $\triangle OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$<br>Area of parallelogram $=\lvert\mathbf{a}\times\mathbf{b}\rvert$ |
| Geometrical meaning |  <br>$\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ = length of projection of $\mathbf{a}$ onto $\mathbf{b}$ |  <br>perpendicular distance from $A$ to line along $\mathbf{b}$ $\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$<br>area of parallelogram, sides $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$<br>2× area of $\triangle OAB$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$ |
| Note | $\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law | no cancellation law<br>$(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is a vector, but triple cross products are usually not tested |

**Note**

Projection of $\mathbf{a}$ onto $\mathbf{b}$ $\neq$ projection of $\mathbf{b}$ onto $\mathbf{a}$. Perpendicular distance from $A$ to $OB$ $\neq$ from $B$ to $OA$. Triangles without $O$ as a vertex: $\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$.

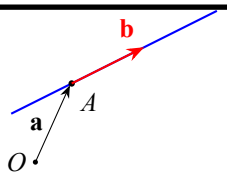
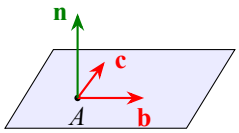
**Useful identity**

For any vectors $\mathbf{a}$, $\mathbf{b}$: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

$\because \mathbf{a}\times\mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$.

# 2 Vectors II

## 2.1 Equations of Lines and Planes

| | Vector | Cartesian | What you need |
|---|---|---|---|
| **Line** |  <br>$\mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$<br>$(\mathbf{r}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$<br>(same line: $\mathbf{r}-\mathbf{a}$ // $\mathbf{b}$) | $\frac{x-a}{d}=\frac{y-b}{e}=\frac{z-c}{f}$<br>$\mathbf{b}=\begin{pmatrix}d\\e\\f\end{pmatrix}$; if $d=0$: $x=a$ separately<br>all coefficients of $x$, $y$ and $z$ in the numerators must be 1 | **Need:** 1 point + 1 direction<br>or 2 points $A$, $B$:<br>$\mathbf{b}=\overrightarrow{AB}$ |
| **Plane** |  <br>$\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$<br>$\mathbf{r}\cdot\mathbf{n}=d$<br>(scalar product form) | $ax+by+cz=d$ | **Need:** 1 point + 2 non-parallel directions // plane<br>or 1 point + normal $\mathbf{n}$<br>$\mathbf{n}=\mathbf{b}\times\mathbf{c},\quad d=\mathbf{a}\cdot\mathbf{n}$<br>3 points: $\mathbf{b}=\overrightarrow{AB}$, $\mathbf{c}=\overrightarrow{AC}$;<br>contains a line: use its direction |

$\mathbf{a}$: position vector of any point on the line/plane. $\mathbf{b}$, $\mathbf{c}$: any two linearly independent directions parallel to the plane.

The equation of any line is not unique.

**Note**

To check if a given point $A$ lies on

- a line: $\mathbf{a}_A=\mathbf{a}+\lambda\mathbf{b}$ (one $\lambda$ satisfies all 3 components)
- a plane: $\mathbf{a}_A\cdot\mathbf{n}=d$

**Example 2.1**

Line through $(1,2,3)$, $(3,2,2)$

$$\mathbf{b}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$\frac{x-1}{2}=\frac{z-3}{-1},\quad y=2$$

**Example 2.2**

Plane through $B(0,1,1)$, $C(1,1,3)$, $D(2,0,2)$

$$\mathbf{n}=\overrightarrow{BC}\times\overrightarrow{BD}=\begin{pmatrix}1\\0\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\3\\-1\end{pmatrix}$$

$$d=\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\-1\end{pmatrix}=2$$

$$2x+3y-z=2$$

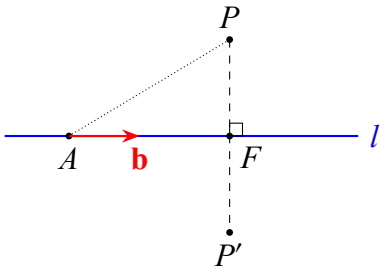
## 2.2 Interactions Map

| | Point | Line | Plane |
|---|---|---|---|
| **Point** | distance $\lvert\overrightarrow{AB}\rvert$ | | |
| **Line** | • on line? §2.1<br>• foot $F$ §2.3<br>• shortest distance §2.3<br>• reflection §2.3 | • // / intersect / skew §2.4<br>• point of intersection §2.4<br>• angle §2.6<br>• reflect line in line §2.7 | |
| **Plane** | • on plane? §2.1<br>• foot $F$ §2.3<br>• shortest distance §2.3<br>• reflection §2.3 | • // / lies in / intersect §2.4<br>• point of intersection §2.4<br>• angle §2.6<br>• reflect line in plane §2.7 | • // / same / intersect §2.4<br>• line of intersection §2.4<br>• angle §2.6<br>• distance (//) §2.8 |

Always draw the diagram first.

## 2.3 Foot / Distance / Reflection

**Point $P$ – Line $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$**

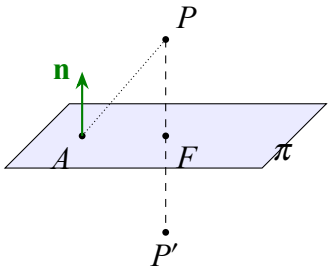



1. $\overrightarrow{OF}=\mathbf{a}+\lambda\mathbf{b}$ for some $\lambda\in\mathbb{R}$
2. $\overrightarrow{PF}\cdot\mathbf{b}=0 \Rightarrow$ solve for $\lambda$
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

**Distance**

$$PF=\lvert\overrightarrow{AP}\times\hat{\mathbf{b}}\rvert$$

**Point $P$ – Plane $\mathbf{r}\cdot\mathbf{n}=d$**




1. $l_{PF}:\ \mathbf{r}=\mathbf{p}+\lambda\mathbf{n},\ \lambda\in\mathbb{R}$
2. $(\mathbf{p}+\lambda\mathbf{n})\cdot\mathbf{n}=d$ (intersection between line & plane)
   $\Rightarrow$ solve for $\lambda$
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

**Distance**

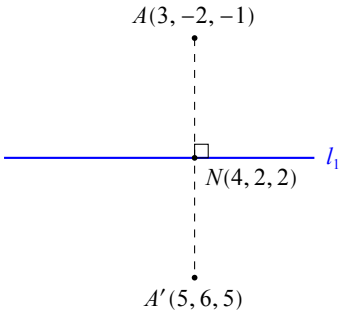
$$PF=\lvert\overrightarrow{AP}\cdot\hat{\mathbf{n}}\rvert$$

## Reflection (and foot of perpendicular) of point to line

**Example 2.3**

Find the foot of the perpendicular $N$ from $A(3,-2,-1)$ to the line $l_1$, and the reflection $A'$ of $A$ in $l_1$, where

$$l_1: \mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}, \quad \lambda\in\mathbb{R}$$




$$\overrightarrow{AN}=\begin{pmatrix}-2+3\lambda\\4\\4-\lambda\end{pmatrix}$$

$$\overrightarrow{AN}\cdot\begin{pmatrix}3\\0\\-1\end{pmatrix}=0$$

$$10\lambda-10=0 \Rightarrow \lambda=1$$

$$\therefore N=(4,2,2)$$

$$\overrightarrow{OA'}=2\begin{pmatrix}4\\2\\2\end{pmatrix}-\begin{pmatrix}3\\-2\\-1\end{pmatrix}$$

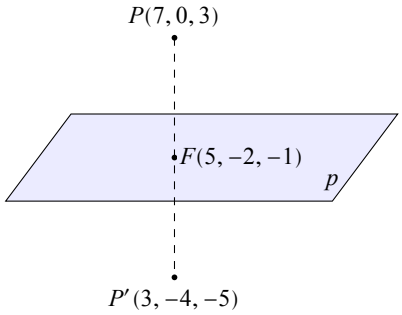
$$\therefore A'=(5,6,5)$$

## Reflection (and foot of perpendicular) of point to plane

**Example 2.4**

Find the foot of the perpendicular $F$ from $P(7,0,3)$ to the plane $p$, and the reflection $P'$ of $P$ in $p$, where

$$p: \mathbf{r}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$




$$\left(\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$

$$13+6\lambda=1 \Rightarrow \lambda=-2$$

$$\therefore F=(5,-2,-1)$$

$$\overrightarrow{OP'}=2\begin{pmatrix}5\\-2\\-1\end{pmatrix}-\begin{pmatrix}7\\0\\3\end{pmatrix}$$

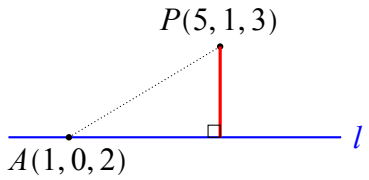
$$\therefore P'=(3,-4,-5)$$

### Shortest distance from point to line

**Example 2.5**

Find the shortest distance from $P(5,1,3)$ to the line

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$




$$\text{shortest distance} = |\overrightarrow{AP} \times \hat{\mathbf{b}}|$$

$$= \left| \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \times \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right|$$

$$= \left| \frac{1}{3} \begin{pmatrix} 3 \\ -6 \\ -6 \end{pmatrix} \right| = \left| \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} \right|$$

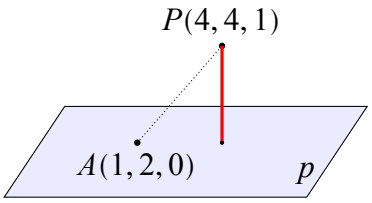
$$= 3$$

### Shortest distance from point to plane

**Example 2.6**

Find the shortest distance from $P(4,4,1)$ to the plane

$$p: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 4$$




$$\text{shortest distance} = |\overrightarrow{AP} \cdot \hat{\mathbf{n}}|$$

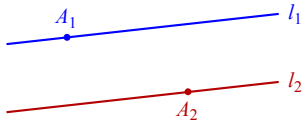
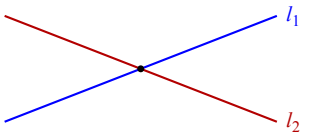
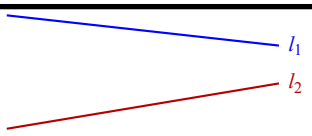
$$= \left| \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right|$$

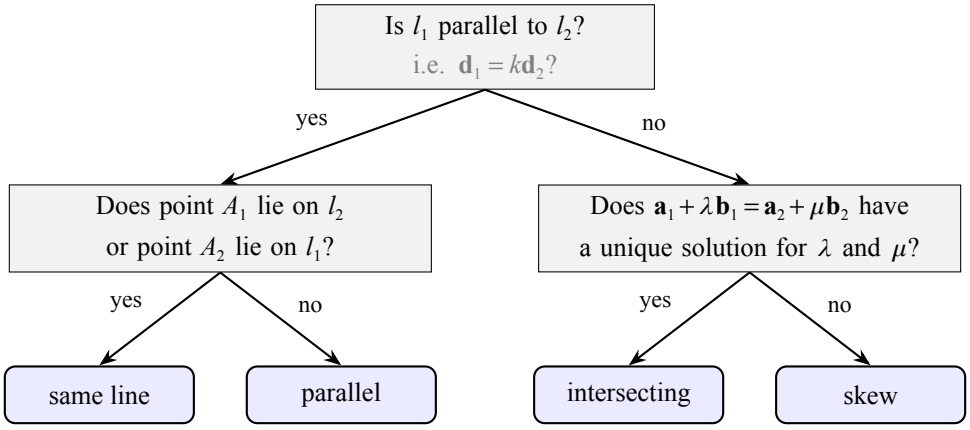
$$= \left| \frac{1}{3}(6+2-2) \right|$$

$$= 2$$

## 2.4 Relationships and Intersections

### Line – Line $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{d}_1$, $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{d}_2$

| | Parallel | Intersecting | Skew |
|---|---|---|---|
| **Diagram** |  $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$<br>$l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$ |  $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$<br>$l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$ |  $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$<br>$l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$ |
| **Condition** | $\mathbf{d}_1 \parallel \mathbf{d}_2$ and $A_1 \notin l_2$, $A_2 \notin l_1$ | $\mathbf{d}_1 \nparallel \mathbf{d}_2$, unique solution for $\lambda$ and $\mu$ | $\mathbf{d}_1 \nparallel \mathbf{d}_2$, no solution for $\lambda$ and $\mu$ |




**Note**

Definition of **skew lines**:

- non parallel
- non intersecting

Thus, to prove that two lines are skew, we must first state that $\mathbf{d}_1 \neq k\mathbf{d}_2$, so $l_1$ and $l_2$ are not parallel, before proceeding with simultaneous equations to show they have no solution.

**Example 2.7**

Determine whether $l_1$ and $l_2$ are intersecting or skew lines.

$$l_1: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$l_2: \mathbf{r}=\begin{pmatrix}2\\12\\5\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\0\end{pmatrix},\quad \mu\in\mathbb{R}$$

$$1+\lambda=2-\mu \qquad —(1)$$

$$2\lambda=12+3\mu \qquad —(2)$$

$$2+\lambda=5 \qquad —(3)$$

From (1) and (3), $\lambda=3,\quad \mu=-2$

Sub into (2), LHS $=6,\quad$ RHS $=12+3(-2)=6$

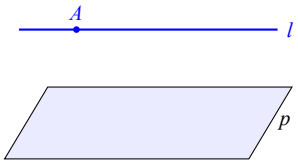
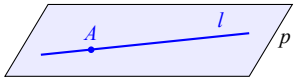
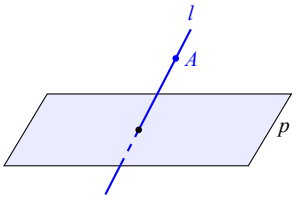
$\therefore$ LHS $=$ RHS

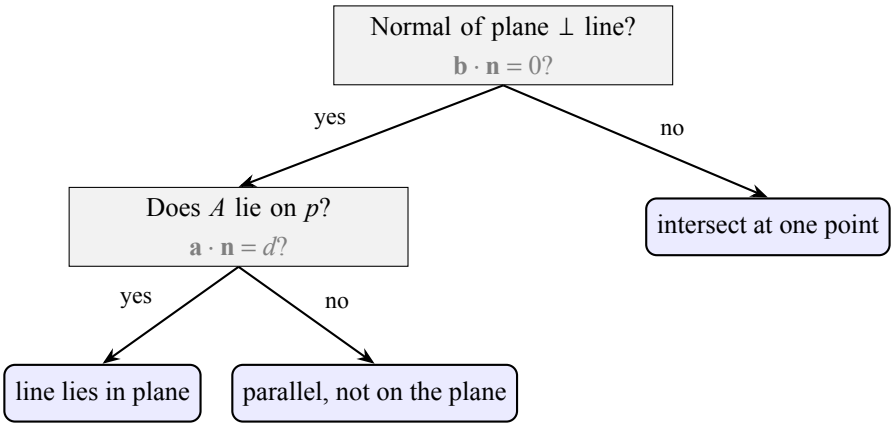
$$\text{Sub } \lambda=3 \text{ into } l_1: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+3\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}4\\6\\5\end{pmatrix}$$

They intersect at $(4, 6, 5)$

Solve 2 of the 3 equations, sub into the 3rd: LHS = RHS ⇒ intersect; LHS ≠ RHS ⇒ skew.

## Line – Plane $\quad l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \pi: \mathbf{r}\cdot\mathbf{n}=d$

| | Parallel, not on the plane | Line lies in the plane | Intersects at one point |
|---|---|---|---|
| **Diagram** |  $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$<br>$p: \mathbf{r}\cdot\mathbf{n}=d$ |  $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$<br>$p: \mathbf{r}\cdot\mathbf{n}=d$ |  $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$<br>$p: \mathbf{r}\cdot\mathbf{n}=d$ |
| **Condition** | • normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$<br>• $A$ does not lie on $p$: $\mathbf{a}\cdot\mathbf{n}\neq d$ | • normal $\perp$ line: $\mathbf{b}\cdot\mathbf{n}=0$<br>• $A$ lies on $p$: $\mathbf{a}\cdot\mathbf{n}=d$ | • normal not $\perp$ line: $\mathbf{b}\cdot\mathbf{n}\neq 0$ |




**Example 2.8**

Find the point of intersection between the line $l$ and plane $p$.

$$l: \mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$p: \mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 5$$

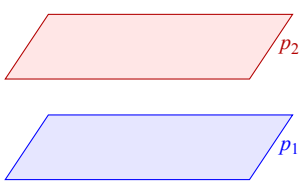
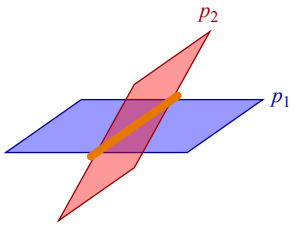
$$\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+\lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

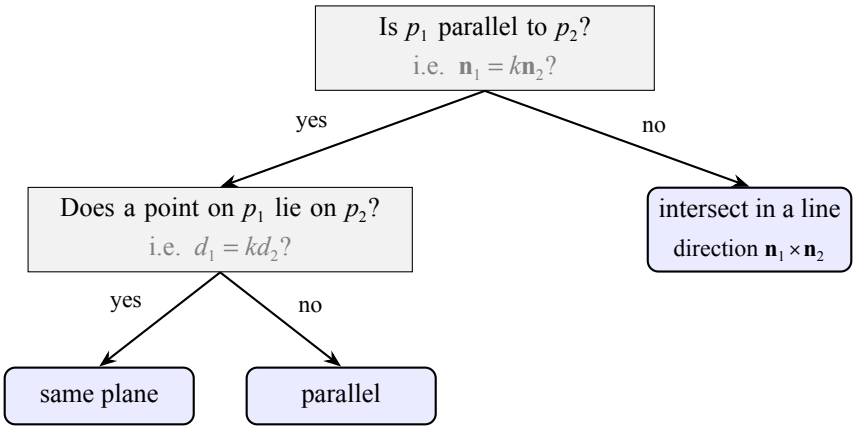
$$\begin{pmatrix} 1+\lambda \\ 3-2\lambda \\ 1+\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 5$$

$$8 - 2\lambda = 5 \Rightarrow \lambda = \tfrac{3}{2}$$

$$\therefore (2.5, 0, 2.5)$$

## Plane – Plane $p_1 : \mathbf{r} \cdot \mathbf{n}_1 = d_1,\ p_2 : \mathbf{r} \cdot \mathbf{n}_2 = d_2$

| | Parallel | Intersecting in a line |
|---|---|---|
| **Diagram** |   |   |
| **Condition** | $\mathbf{n}_1 \parallel \mathbf{n}_2, p_1 \neq p_2$ | $\mathbf{n}_1 \nparallel \mathbf{n}_2$ |




**Example 2.9**

Find the intersection between the planes:

$$3x+2y+z=10$$

$$x-2y+z=5$$

$$\text{GC: } x=\frac{15}{4}-\frac{1}{2}z$$

$$y=-\frac{5}{8}+\frac{1}{4}z$$

$$z=z$$

$$\mathbf{r}=\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}\frac{15}{4}-\frac{1}{2}z\\-\frac{5}{8}+\frac{1}{4}z\\z\end{pmatrix}$$

$$=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+z\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix}$$

$$\therefore l:\ \mathbf{r}=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+z\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix},\quad z\in\mathbb{R}$$

$\mathbf{d}=\mathbf{n}_1\times\mathbf{n}_2=(4,-2,-8)^{\mathrm{T}}\,//\,(2,-1,-4)^{\mathrm{T}}$ checks the direction.

**Example 2.10**

Sometimes, we may not be able to use our GC to find the line of intersection between two planes. The steps to find it manually are:

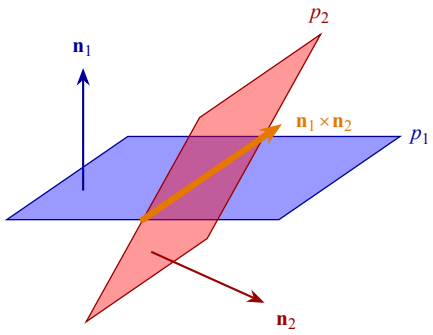
$$p_1: \mathbf{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=6, \qquad p_2: \mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=2$$

**Step 1:** direction $=\mathbf{n}_1\times\mathbf{n}_2$

$$\mathbf{d}=\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$=\begin{pmatrix}1(2)-1(-1)\\1(1)-1(2)\\1(-1)-1(1)\end{pmatrix}$$

$$=\begin{pmatrix}3\\-1\\-2\end{pmatrix}$$




**Step 2:** find a point that lies on both planes

$$p_1: x+y+z=6$$

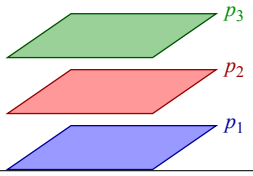
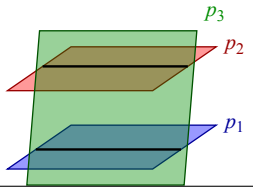
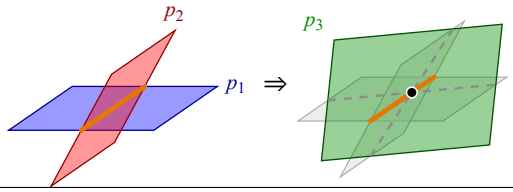
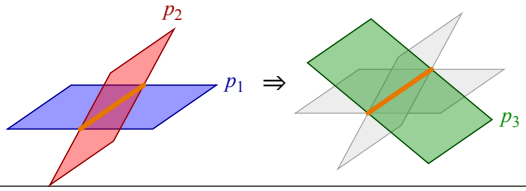
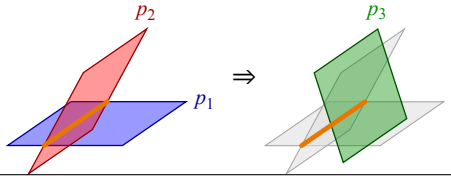
$$p_2: x-y+2z=2$$

$$\text{Let } z=0$$

$$\left.\begin{aligned}x+y&=6\\x-y&=2\end{aligned}\right\}\Rightarrow\begin{aligned}x&=4\\y&=2\end{aligned}$$

$$\therefore l: \mathbf{r}=\begin{pmatrix}4\\2\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\-1\\-2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

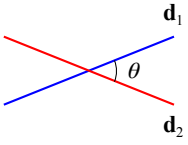
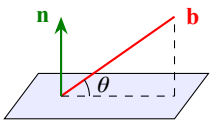
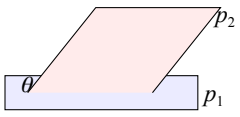
## 2.5 Interaction between 3 Planes

| Case | Diagram(s) | Condition |
|---|---|---|
| **1. All parallel** |  | — |
| **2. Two parallel, one not** |  | — |
| **3. All intersect at 1 point** (most common) |  | $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ intersects $p_3$ (see p. 8) |
| **4. All intersect in a line** |  | $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ lies on $p_3$ (see p. 8) |
| **5. Form a triangular prism** (no single point of intersection) |  | $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ does not lie on $p_3$ (see p. 8) |

**Note**

To analyse the interaction between 3 arbitrary planes, we can study how the **line of intersection between $p_1$ and $p_2$** interacts with $p_3$.

## 2.6 Angles

| Line – Line | Line – Plane | Plane – Plane |
|---|---|---|
|  |  |  |
| $\cos\theta=\dfrac{\lvert\mathbf{d}_1\cdot\mathbf{d}_2\rvert}{\lvert\mathbf{d}_1\rvert\lvert\mathbf{d}_2\rvert}$ | $\sin\theta=\left\lvert\dfrac{\mathbf{n}\cdot\mathbf{b}}{\lvert\mathbf{n}\rvert\lvert\mathbf{b}\rvert}\right\rvert$ | $\cos\theta=\left\lvert\dfrac{\mathbf{n}_1\cdot\mathbf{n}_2}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}\right\rvert$ |
| directions | sin: $\angle(\mathbf{n},\mathbf{b})=90^\circ-\theta$ | normals |

$\theta$ = acute angle.

**Example 2.11**

$3x-y+2z=3,\ y=2$

$$\cos\theta=\frac{|-1|}{\sqrt{14}\cdot 1}$$

$$\theta=74.5^\circ$$

**Example 2.12**

$$\mathbf{r}=\begin{pmatrix}-2\\-4\\5\end{pmatrix}+\mu\begin{pmatrix}3\\5\\-4\end{pmatrix},\ \mu\in\mathbb{R},\ \mathbf{r}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}=1$$

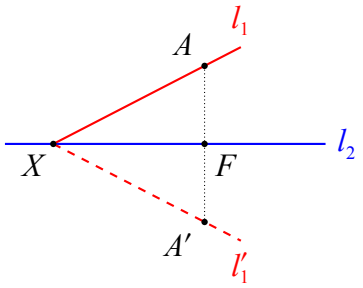
$$\sin\theta=\frac{5}{\sqrt{50}\cdot 1}=\frac{1}{\sqrt{2}}$$

$$\theta=45^\circ$$

**DRAFT:** Your source lists 4.6 Angle Involving Lines, but that page is not in the PDF. The line–line formula above is the standard one; please confirm.

## 2.7 Reflection of a Line

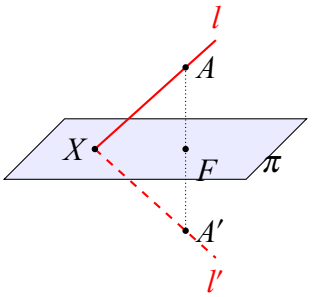
**Line $l_1$ in line $l_2$** (intersect at $X$)




1. $X=l_1\cap l_2$
2. $A\in l_1$: foot $F$ on $l_2$
3. $\overrightarrow{OA'}=2\overrightarrow{OF}-\overrightarrow{OA}$
4. $l_1'$: $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

If $l\ //\ \pi$: $l'$ has the same direction, through $A'$.

**Line $l$ in plane $\pi$** (intersect at $X$)




1. $X=l\cap\pi$
2. $A\in l$: foot $F$ on $\pi$
3. $\overrightarrow{OA'}=2\overrightarrow{OF}-\overrightarrow{OA}$
4. $l'$: $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

**Example 2.13**

$l:\mathbf{r}=(7\mathbf{i}+3\mathbf{k})+\lambda(2\mathbf{i}+\mathbf{k}),\ \lambda\in\mathbb{R},\ \pi:\mathbf{r}\cdot(\mathbf{i}+\mathbf{j}+2\mathbf{k})=1$

$$(7+2\lambda)+2(3+\lambda)=1\ \Rightarrow\ \lambda=-3$$

$$\therefore X=(1,0,0)$$

$$A(7,0,3):\ F=(5,-2,-1),\ A'=(3,-4,-5)\quad(\S2.3)$$

$$\overrightarrow{XA'}=\begin{pmatrix}2\\-4\\-5\end{pmatrix}$$

$$\therefore l':\ \mathbf{r}=\mathbf{i}+\mu(2\mathbf{i}-4\mathbf{j}-5\mathbf{k}),\ \mu\in\mathbb{R}$$

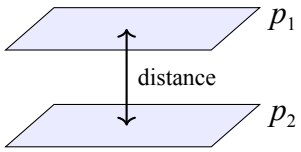
**DRAFT:** 4.7 Reflection (Line about Line) is listed in your source but its page is missing; the steps on the left are the standard method. Add an example from an official answer sheet?

## 2.8 Distance Between Parallel Planes

$$p_1 : \mathbf{r}\cdot\mathbf{n}=d_1, \qquad p_2 : \mathbf{r}\cdot\mathbf{n}=d_2$$

**Distance**

$$\text{distance}=\left|\frac{d_1}{|\mathbf{n}|}-\frac{d_2}{|\mathbf{n}|}\right| \qquad \text{plane to } O=\frac{|k|}{|\mathbf{n}|}$$




Same $\mathbf{n}$ for both planes first. $\mathbf{r}\cdot\mathbf{n}=0$: plane through $O$.

**Example 2.14**

$x+2y-3z=2,\ 2x+4y-6z=-10$

$$2x+4y-6z=-10 \Rightarrow x+2y-3z=-5$$

$$\text{distance}=\left|\frac{2}{\sqrt{14}}-\frac{-5}{\sqrt{14}}\right|=\frac{7}{\sqrt{14}}$$